

Raw LiDAR Intensity — Lane Visibility Analysis

66 sequences · 4,901 lane-bearing frames · forward-facing LiDAR · raw intensity, no preprocessing

Core Takeaway

Raw LiDAR intensity is useful but not sufficient on its own. It separates lane markings from asphalt — especially close to the sensor — but it does **not** reliably distinguish lane markings from other painted road markings (stop lines, other paint), because their intensities largely overlap.

Decision: keep intensity as an input feature, but do not treat it as a lane detector on its own. The model needs geometry, local context, and lane-aware sampling.

1. Lane is brighter than road globally

Lane markings are typically **+5 intensity units** brighter than road/asphalt at the median.

Class	Raw ID	Median	Mean	IQR
Lane line marking	8	33	37.77	19
Road	7	28	26.36	6

2. Lane is NOT separable from other paint by intensity

Lane, stop-line, and other-road-marking medians are all ≈ 33 . Intensity helps with paint-vs-asphalt — not lane-paint-vs-other-paint.

Class	Raw ID	Median
Lane line marking	8	33
Stop line marking	9	33
Other road marking	10	33

3. Some sequences have a strong intensity signal

Lane visibility is not uniform across the dataset. A handful of sequences are very intensity-friendly (AUC > 0.8 vs road).

Sequence	Lane points	Median shift	AUC vs road	AUC vs non-lane surface
023	49,342	+23	0.858	0.858
042	20,066	+17	0.830	0.826
053	47,695	+6	0.827	0.818
109	40,057	+17	0.825	0.826
016	118,863	+11	0.817	0.816

4. Best-case frames are extremely clean

Some individual frames show almost perfect intensity-based separation. Example: **sequence 103, frame 50** — lane median 69.5 vs road median 29 (shift +40.5), AUC 0.984. Useful for sanity checks and visual inspection.

5. Local contrast is strong (radius 0.5 m)

The useful intensity signal is largely **local** — lane points stand out against their immediate neighborhood far more than against global statistics.

Metric	Value	Meaning
Δ mean (avg)	+12.46	Lane brighter than local mean
Δ median (avg)	+12.56	Lane brighter than local median
Frac $\geq$ local mean	0.816	81.6% of lane points stand out
Frac $\geq$ local P90	0.598	59.8% beat the local 90th percentile

Implication: a neighborhood-based point-cloud model should benefit — but training patches must actually contain lane neighborhoods.

6. Distance weakens the signal

Raw intensity is mostly useful within ~20 m. Beyond that, lane evidence becomes sparse and the signal collapses toward random.

Distance	Lane points	Lane median	Road median	Shift	AUC
0–10 m	1,013,079	40	29	+11	0.780
10–20 m	14,925	14	7	+7	0.660
20–30 m	421	1	0	+1	0.593
30–40 m	13	0	0	0	0.507

7. Intensity alone is weak for the full road-paint problem

Task	ROC AUC	PR AUC	Best F1
Lane vs road	0.705	0.768	0.667
Lane vs road + other markings	0.585	0.345	0.406

Moderate against asphalt only. Once stop lines and other markings enter the picture, raw intensity alone is not enough.

Caveats

- This measures the raw intensity signal — not final segmentation quality.
- Stage 5 numbers are diagnostic, not a final classifier benchmark.
- The next modeling step must account for geometry, local context, sparse far-range lane points, and confusion with other road markings.